

Regression Problem Set

① Scatter diagram and linear relationship.

X (Hours studied)

Y (Exam Score)

1

2

2

3

3

4

4

5

5

6

6

7

Q ② Comment on whether a linear relationships seems appropriate between X & Y

A.2. There appears linear relationship as study hours increase marks also increase. Follow straight line pattern, indicates positive linear relationship.

② Model Representation for Single Variable Regression

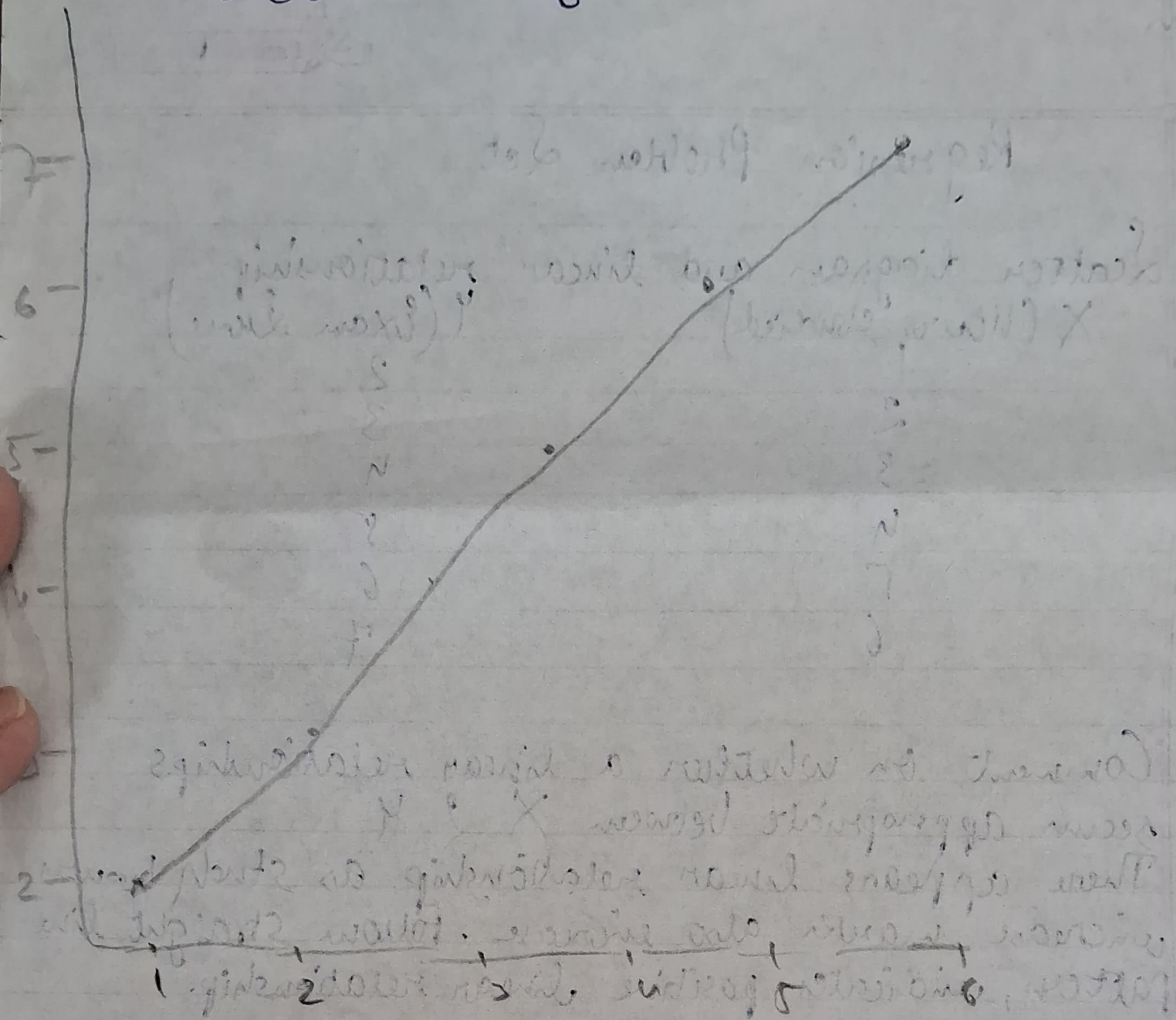
$$\hat{y} = b_0 + b_1 x + b$$

b: Expected exam Score if student studied x hours

m: Increase in Exam Score for each additional hour studied.

$$\rightarrow \hat{y} = 1 + x$$

Scatter diagram



Model representation for high variable regression

$$\hat{y} = a + bx$$
 where x is the independent variable and y is the dependent variable.
 The regression line is the line of best fit.

$$x + 1 = y$$

③ Write Mean Squared Error (MSE) cost function

$$J(B_0, B_1) = \frac{1}{n} \sum_{i=1}^n (y_i - (\beta_0 + \beta_1 x_i))^2$$

y_i : actual value $\hat{y}_i$: predicted value

② Computing cost for given Parameter.

x	y	$\hat{y}$	$(y - \hat{y})^2$
1	2	0	4
2	3	0	9
3	4	0	16
4	5	0	25
5	6	0	36
6	7	0	49

$\sum (y - \hat{y})^2 = 139$

$= \frac{139}{6} = 23.17$

(ii) $B_0 = 0, B_1 = 1$ $\hat{y} = m \cdot x + c$ $1(x) + 0 = x$

x	y	$\hat{y}$	$(y - \hat{y})^2$
1	2	1	1
2	3	2	1
3	4	3	1
4	5	4	1
5	6	5	1
6	7	6	1

$6/6 = 1$

(iii) $B_0 = 2$

$B_1 = 0.5$

$$\hat{y} = 2 + 0.5x$$

X	y	$\hat{y}$	$(y - \hat{y})^2$
1	2	2.5	0.25
2	3	3	0
3	4	3.5	0.25
4	5	4	1
5	6	4.5	2.25
6	7	5	4

$$\sum (y - \hat{y})^2 = 7.75$$

$$7.75 / 6 = 1.29$$

③ Derive partial derivatives $\frac{\partial J}{\partial \beta_0}$ and $\frac{\partial J}{\partial \beta_1}$

$$J(\beta_0, \beta_1) = \frac{1}{n} \sum (y_i - (\beta_0 + \beta_1 x_i))^2$$

$$\frac{\partial J}{\partial \beta_0} = \frac{1}{n} \sum \frac{\partial}{\partial \beta_0} (y_i - (\beta_0 + \beta_1 x_i))^2$$

Applying chain Rule

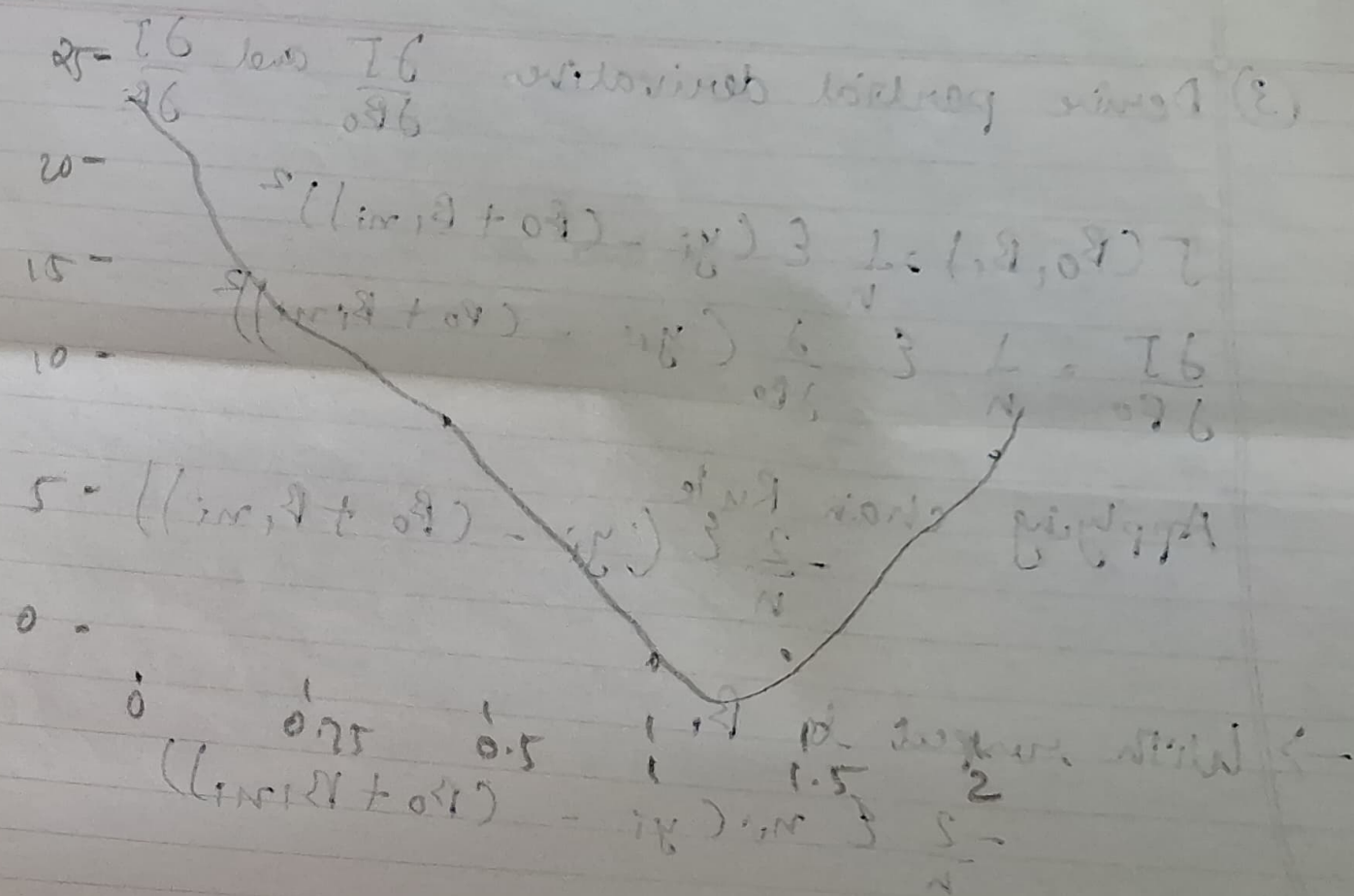
$$= \frac{-2}{n} \sum (y_i - (\beta_0 + \beta_1 x_i))$$

→ With respect to β_1

$$= \frac{-2}{n} \sum x_i (y_i - (\beta_0 + \beta_1 x_i))$$

④ (Visualization) Fix $\beta_0 = 0$ and plot $J(0, \beta_1)$ as a function of $\beta_1 \in [0, 2]$ to illustrate convexity of the cost. Error²

Ans	$\beta_1 = 0$	$\hat{y} = 0$	$J(0, 0) = 23.17$
	$\beta_1 = 0.5$	$\hat{y} = 0.5x$	8.29
	$\beta_1 = 1$	$\hat{y} = x$	(minimum)
	$\beta_1 = 1.5$	$\hat{y} = 1.5x$	1.29
	$\beta_1 = 2$	$\hat{y} = 2x$	9.17



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- $f(0,0) = 0$
- $f(1,0) = 0$
- $f(2,0) = 0$
- $f(0,1) = 0$
- $f(1,1) = 0$
- $f(2,1) = 0$
- $f(0,2) = 0$
- $f(1,2) = 0$
- $f(2,2) = 0$

Q.4(i) Write formulas least squares

$$B_1 = \frac{\sum (X_i - \bar{x})(Y_i - \bar{y})}{\sum (X_i - \bar{x})^2} \quad B_0 = \bar{y} - B_1 \bar{x}$$

② Compute m and b

X	y	$X - \bar{x}$	$y - \bar{y}$	$(X - \bar{x})(y - \bar{y})$	$(X - \bar{x})^2$
1	2	-2.5	-2.5	6.25	6.25
2	3	-1.5	-1.5	2.25	2.25
3	4	-0.5	-0.5	0.25	0.25
4	5	0.5	0.5	0.25	0.25
5	6	1.5	1.5	2.25	2.25
6	7	2.5	2.5	6.25	6.25
$\bar{x}$	$\bar{y}$				
3.5	4.5				

$$B_1 = 17.5 / 17.5 = 1$$

$$B_0 = \bar{y} - B_1 \bar{x} = 4.5 - (1)(3.5) = 1$$

$$\hat{y} = mx + c = x + 1$$

$\hat{y} = mx + c$ The linear regression line plotted.

⑤ Simple Linear Regression + Gradient Descent

	X (Hours)	$x - \bar{x}$	Y (Score)	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
	1	-2	2	-2	4	4
	2	-1	4	0	0	1
$\frac{15}{5} = 3$	3	0	5	1	0	0
	4	1	4	0	0	1
$\frac{20}{5} = 4$	5	2	5	1	2	4
	$\frac{15}{5}$		$\frac{20}{5}$		$\frac{6}{5}$	$\frac{10}{5}$

$$(m) \beta_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{6/10}{10/5} = 0.6$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 4 - (0.6) \cdot 3 = 4 - 1.8 = 2.2$$

$$\hat{y} = \beta_0 + \beta_1 x \text{ (m x 1b)} = 0.6x + 2.2$$

at $x = 6$

$$= 0.6(6) + 2.2 = 2.2 + 3.6 = 5.8$$

→ For Gradient descent

$$\begin{aligned} \hat{y} - y &= -2 \\ &-4 \\ &-5 \\ &-4 \\ &-5 \end{aligned} \quad \begin{aligned} \sum (\hat{y} - y) &= -20 \\ \sum (\hat{y} - y)x &= -66 \end{aligned} \quad \begin{aligned} \beta_0 &= 0 \quad \beta_1 = 1 \quad \alpha = 0.1 \\ (\hat{y} - y)x &= -2 \\ &-8 \\ &-10 \\ &-10 \\ &-10 \end{aligned}$$

→ Updated B_0

$$B_0^{\text{new}} = 2.2 - 0.1 \left(\frac{-20}{5} \right) = 2.6$$

$\boxed{2.6}$

Updated B_1

$$B_1^{\text{new}} = 0.6 - 0.1 \left(\frac{-66}{5} \right) = 1.92$$

$\boxed{1.92}$

$\boxed{B_0 = 2.6}$ $\boxed{B_1 = 1.92}$

⑥ Gradient descent for Single Variable Linear Regression

X	Y	$\hat{y}$	$\hat{y} - y$	$(\hat{y} - y)X$
1	1.5	0	-1.5	-1.5
2	1.9	0	-1.9	-3.8
3	3.2	0	-3.2	-9.6
4	3.0	0	-3	-12
5	4.1	0	-4.1	-20.5

→ Derive Cost function update rules

$$\frac{\partial J}{\partial B_0} = \frac{1}{m} \sum (\hat{y} - y) = B_0 - B_0 - 2 \frac{1}{m} \sum (\hat{y} - y)$$

$$\frac{\partial J}{\partial B_1} = \frac{1}{m} \sum (\hat{y} - y) X_i = B_1 - B_1 - 2 \frac{1}{m} \sum (\hat{y} - y) X_i$$

$$\sum (y^A - y) = -13.7$$

$$\sum (y^A - y)x = -47.4$$

$$\text{Updated } B_0 = 0 - 0.1 \left(\frac{-13.7}{5} \right) = 0.274$$

$$B_1 = 0 - 0.1 \left(\frac{-47.4}{5} \right) = 0.948$$

$$\hat{y} = 0.274 + 0.948x$$

x	y	$\hat{y}$	$\hat{y} - y$	$(\hat{y} - y)x$
1	1.5	1.222	-0.278	-0.278
2	1.9	2.17	0.27	0.540
3	3.2	3.118	-0.082	-0.246
4	3	4.066	1.066	4.264
5	4.1	5.014	0.914	4.570
				<u>8.85</u>

$$\sum (y^A - y) = 1.89$$

$$\begin{aligned} \text{Update } B_0 &= 0.274 - 0.1 (1.89/5) \\ &= 0.274 - 0.1 (0.378) = 0.2362 \\ &\quad \boxed{0.24} \end{aligned}$$

$$\text{Update } B_1 = 0.948 - 0.1 \left(\frac{8.85}{5} \right)$$

$$\begin{aligned} &0.948 - 0.1 (1.77) \\ &\quad \boxed{0.771} \end{aligned}$$

$$S(\hat{y}) = (y - \hat{y})^T (I - H)^{-1} (y - \hat{y})$$

$$= (y - \hat{y})^T (I - H)^{-1} (y - \hat{y})$$

$$= (y - \hat{y})^T (I - H)^{-1} (y - \hat{y})$$

$y - \hat{y}$	$y - \hat{y}$	$\hat{y}$	$\hat{y}$	χ^2
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0
0.0	0.0	0.0	0.0	0

$$S(\hat{y}) = (y - \hat{y})^T (I - H)^{-1} (y - \hat{y})$$

$$= (y - \hat{y})^T (I - H)^{-1} (y - \hat{y})$$

$$= (y - \hat{y})^T (I - H)^{-1} (y - \hat{y})$$

⑦ Linear Regression via Normal Equation
Want to predict final exam score

X (Hours) Y (Score)

1

2

3

4

① Linear Regression model
 $y = \beta_0 + \beta_1 x$

② Matrix Form
add column 1s for intercept

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

Matrix Y:

$$\begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

Parameter Vals

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

→ Normal Equation = $(X^T X)^{-1} X^T Y$

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \times X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

8

$(1 \times 1) + (1 \times 1) + (1 \times 1) = 3$ (4)

$(1 \times 1) + (1 \times 2) + (1 \times 3) = 6$

$(1 \times 1) + (2 \times 1) + (3 \times 1) = 6$

$(1 \times 1) + (2 \times 2) + (3 \times 3) = 14$

Linear Regression model
 $y = \beta_0 + \beta_1 x$

Matrix form
 for intercept

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}$$

Matrix $Y = \begin{bmatrix} 3 \\ 6 \\ 6 \end{bmatrix}$

Normal equation: $(X^T X)^{-1} X^T Y$

$$X^T = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix}$$

$$X^T X = \begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix}$$

$$(X^T X)^{-1} = \frac{1}{\det \begin{bmatrix} a & -b \\ -c & a \end{bmatrix}}$$

$$\det : (3 \times 14) - (6 \times 6) = 42 - 36 = 6$$

$$(X^T X)^{-1} = \frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} =$$

$$X^T Y = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \end{bmatrix} \begin{bmatrix} 2 \\ 4 \\ 6 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 12 \\ 28 \end{bmatrix} \quad \begin{matrix} (1 \times 2) + (1 \times 4) + (1 \times 6) \\ (1 \times 2) + (2 \times 4) + (3 \times 6) \end{matrix}$$

$$(X^T X)^{-1} X^T Y$$

$$\frac{1}{6} \begin{bmatrix} 14 & -6 \\ -6 & 3 \end{bmatrix} \begin{bmatrix} 12 \\ 28 \end{bmatrix}$$

$$\frac{1}{6} \begin{bmatrix} 0 \\ 12 \end{bmatrix} \quad \begin{matrix} (14 \times 12) + (-6 \times 28) = 168 - 168 = 0 \\ (-6 \times 12) + (28 \times 3) = -72 + 84 = 12 \end{matrix}$$

$$\begin{matrix} 0 & 16 \\ 12 & 16 \end{matrix} = \begin{bmatrix} 0 \\ 2 \end{bmatrix} \quad B_0 = 0 \quad B_1 = 2$$

$$y = 0 + 2x \quad y = 2x$$

$$\text{at } x = 4$$

$$= 2(4) = 8$$

Q.8 Linearity Assumption.

X	Y	change in Y
1	5	-
2	9	4
3	15	6
4	25	10
5	40	15

Increase is not constant.
Growth becomes faster as X increases.
So data curves upward. Relationship is not linear.

→ Possible Transformation of X
To make more linear.

① Log Transformation

$$X' = \log(X)$$

$$Y = \beta_0 + \beta_1 \log(X)$$

② Polynomial Transform

Y grows rapidly

$$X' = X^2$$

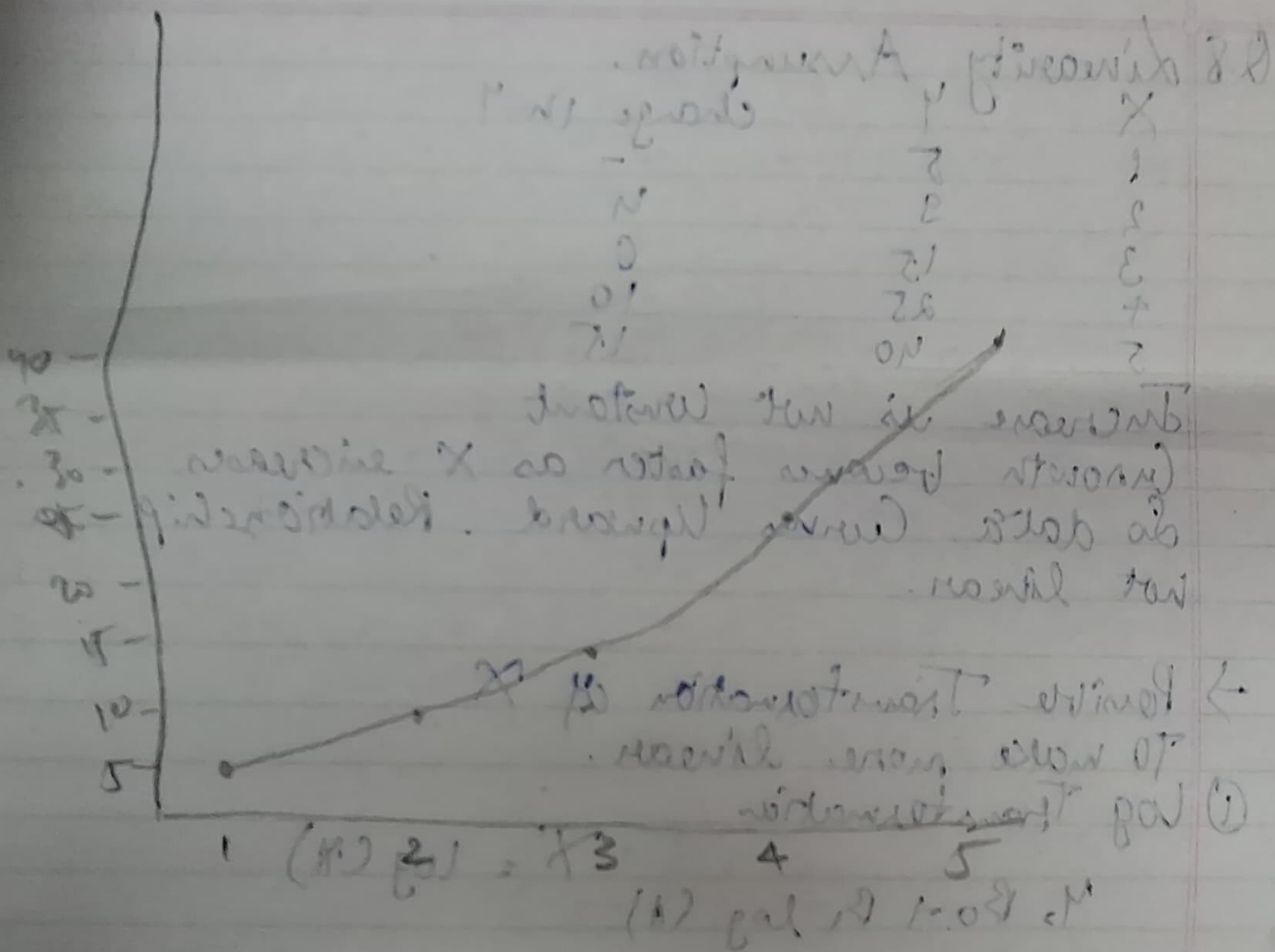
$$Y = \beta_0 + \beta_1 X + \beta_2 X^2$$

③ Exponential model (If growth is very fast)

$$Y = ae^{bx}$$

$$\log(XY) = \beta_0 + \beta_1 X$$

Lib



(2) Polymers of the same type
 $X_1 = X_2$

(3) Exponential model (if growth is very fast)
 $X_1 = X_2$
 $X_1 + X_2 = 100\%$

Q.9 Homoscedasticity Assumption
Residuals $\{-2, -1, 0, 1, 5\}$
 $Y = \{30, 40, 50, 60, 70\}$

- Residuals measure how far predicted value is from actual value.
e.g. $4 - 4$
- Homoscedasticity means: Variance of residuals should remain constant for all predicted values. Residuals should have similar spread everywhere.
- If homoscedasticity holds residuals should be randomly scattered around zero.
- In this residuals increase as Salary Increase so creates fan shaped pattern.
- Variance is not constant so Homoscedasticity violated so this situation called heteroscedasticity. Error Variance changes across observations.
- Violation Causes
 - ① Incorrect Standard Errors
 - ② Wrong hypothesis Test
 - ③ Poor Confidence intervals.
 - ④ Unstable Prediction

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10 Properties of Regression line

X (Hours) Y

$\bar{x} = 2.5$
 $\bar{y} = 3.5$

		$x - \bar{x}$	$y - \bar{y}$	$(x - \bar{x})(y - \bar{y})$	$(x - \bar{x})^2$
1	2	-1.5	-1.5	2.25	2.25
2	3	-0.5	-0.5	0.25	0.25
3	5	0.5	1.5	0.75	0.25
4	4	1.5	0.5	0.75	2.25
10	14			4.00	5

$$b_1 = \frac{\sum (x - \bar{x})(y - \bar{y})}{\sum (x - \bar{x})^2} = \frac{4}{5} = 0.8$$

$$b_0 = \bar{y} - b_1 \bar{x} = 3.5 - (0.8)(2.5) = 1.5$$

$$\hat{y} = 1.5 + 0.8x$$

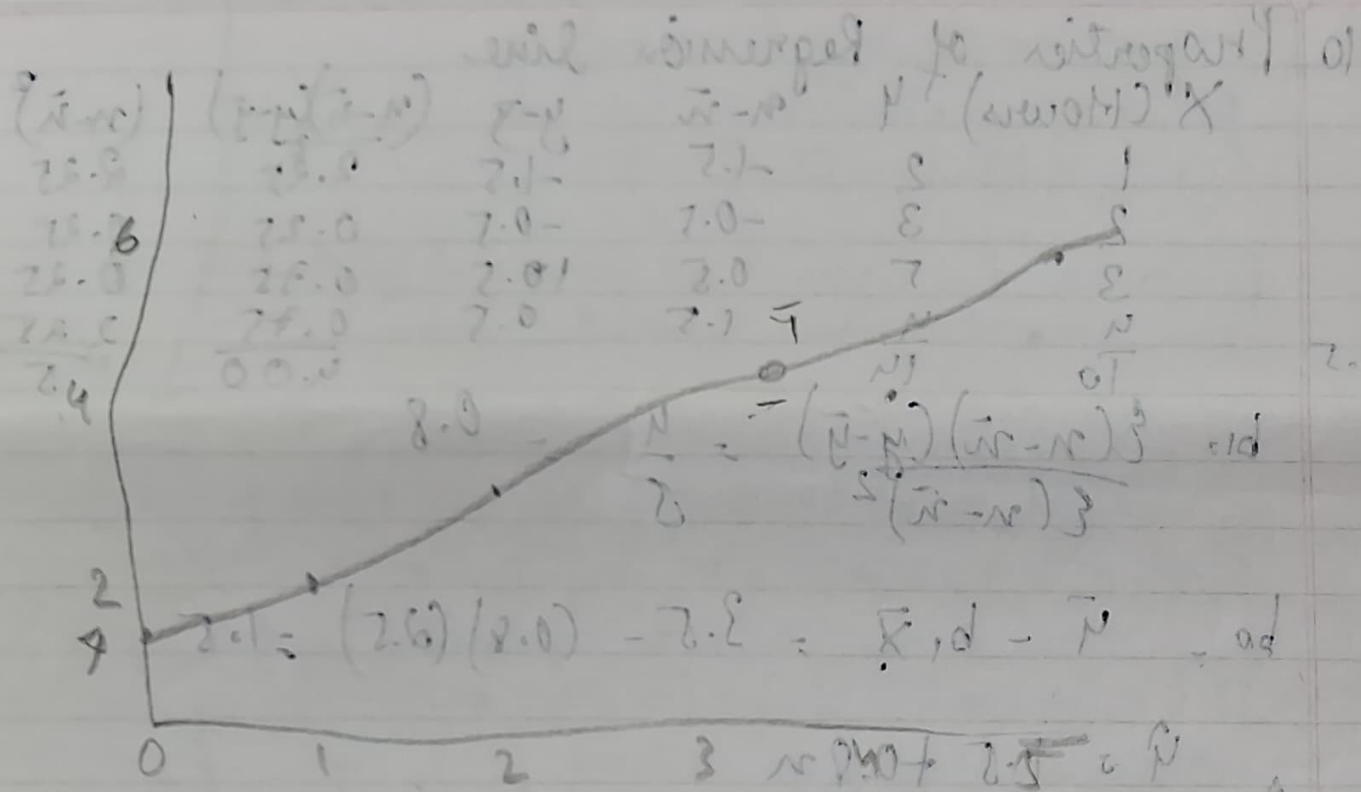
4
2.3
3.1
3.9
4.7

② Verify Regression line passes through mean point
Regression line passes through $(\bar{x}, \bar{y})$

$$\bar{x} = 2.5 \quad \bar{y} = 3.5$$

$$x = 2.5$$

$$1.5 + 0.8(2.5) = 1.5 + 2 = 3.5$$



2.5
2.0
1.5
1.0
0.5
0.0

2) Verify Regression line power forward

$X = 2.2$ $Y = 2.2$

$Y = 0.8X + 1.2$

$2.2 = 0.8(2.2) + 1.2$

$2.2 = 1.76 + 1.2$

$2.2 = 2.96$

→ Compute Residuals.

X	Y	$\hat{Y}$	Residual $y - \hat{y}$
1	2	2.3	-0.3
2	3	3.1	-0.1
3	5	3.9	1.1
4	4	4.7	-0.7

→ Verify Sum of residuals = 0
 $-0.3 - 0.1 + 1.1 - 0.7 = 0$

(11) Model Performance with R^2

$\bar{X} = 2.5$	X	Y	$(n - \bar{n})(y - \bar{y})$	$(n - \bar{n})(y - \bar{y})$	$(n - \bar{n})$
	1	2	-1.5	-1.75	2.25
$\bar{y} = 3.75$	2	4	-0.5	0.25	0.25
	3	5	0.5	1.25	0.25
	4	4	1.5	0.25	2.25
				<u>0.375</u>	<u>5</u>
				1.5	

$$\beta_1 = \frac{\sum (n - \bar{n})(y - \bar{y})}{\sum (n - \bar{n})^2} = \frac{2.5}{5} = 0.7$$

$$\beta_0 = \bar{y} - \beta_1 \bar{x} = 3.75 - (0.7)(2.5) = 2$$

$$\hat{y} = 2 + 0.7x$$

- 2.7
- 3.4
- 4.1
- 4.8

→ Total Sum of Squares (SST) = $\sum (y_i - \bar{y})^2$

y	$y - \bar{y}$	$(y - \bar{y})^2$
2	-1.75	3.0625
4	0.25	0.0625
5	1.25	1.5625
4	0.25	0.0625
Σ		4.75

→ Regression Sum of Squares (SSR) = $\sum (\hat{y} - \bar{y})^2$

$\hat{y}$	$\hat{y} - \bar{y}$	$(\hat{y} - \bar{y})^2$
2.7	-1.05	1.1025
3.4	-0.35	0.1225
4.1	0.35	0.1225
4.8	1.05	1.1025
Σ		2.45

→ Error Sum of Squares. SSE = $\sum (y - \hat{y})^2$

y	$\hat{y}$	Error	Square
2	2.7	-0.7	0.49
4	3.4	0.6	0.36
5	4.1	0.9	0.81
4	4.8	-0.8	0.64
Σ			2.30

$$SST = SSR + SSE$$

$$4.75 = 2.45 + 2.30$$

→ Coefficient of determination R^2

$$R^2 = \frac{SSR}{SST} = \frac{2.45}{4.75} = 0.516$$

→ Interpretation = 0.516

51.6% of variation is explained by regression model using x . So remaining 48.4% unexplained error

(12) Multivariable Regression Model representation

X_1 (Hours)	X_2	y
2	5	70
4	10	88
3	6	78

→ $y = \beta_0 + \beta_1 X_1 + \beta_2 X_2 + e$

$70 = \beta_0 + 2\beta_1 + 5\beta_2 + e$

(2) Matrix Representation $y = X\beta + e$

$$y = \begin{bmatrix} 70 \\ 88 \\ 78 \end{bmatrix}$$

$$\text{Matrix } X = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 4 & 10 \\ 1 & 3 & 6 \end{bmatrix}$$

Parameter Vector $\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$

Intercept
weight for hours
weight for problems.

→ Error e , $\begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$

$$\begin{bmatrix} 70 \\ 88 \\ 71 \end{bmatrix}, \begin{bmatrix} 1 & 2 & 5 \\ 1 & 4 & 10 \\ 1 & 3 & 6 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix} + \begin{bmatrix} e_1 \\ e_2 \\ e_3 \end{bmatrix}$$

For Verifying

$$\begin{bmatrix} \beta_0 + 2\beta_1 + 5\beta_2 \\ \beta_0 + 4\beta_1 + 10\beta_2 \\ \beta_0 + 3\beta_1 + 6\beta_2 \end{bmatrix}$$

Matrix	dimension	Reason
Y	3×1	3 observations
X	3×3	3 rows, 2 features + Inter
β	3×1	3 parameters
e	3×1	error for each

(13) Multivariable Cost Function and Gradient.

X_1	X_2	Y
2	5	70
4	10	88
3	6	78

$$X = \begin{bmatrix} 1 & 2 & 5 \\ 1 & 4 & 10 \\ 1 & 3 & 6 \end{bmatrix}$$

$$Y = \begin{bmatrix} 70 \\ 88 \\ 78 \end{bmatrix}$$

$$\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$$

→ Cost Function (MSE)

$$J(\beta) = \frac{1}{2m} \sum_{i=1}^m (\hat{y} - y)^2$$

$$\frac{1}{6} \|X\beta - y\|^2$$

→ Derive Gradient

$$\nabla_{\beta} J(\beta) = \frac{1}{m} X^T (X\beta - y)$$

→ Evaluate $J(\beta)$ at $\beta = 0$ $\beta = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

$$X\beta = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

Error Vector, $X\beta - y = \begin{bmatrix} -70 \\ -88 \\ -78 \end{bmatrix}$

Square Error: $4900 + 7744 + 6084 = 18728$

$$J(\beta) = \frac{1}{6} (18728)$$

$$\boxed{= 3121.33}$$

→ Compute Gradient $\beta = 0$

$$X^T, \begin{matrix} & X^T Y \\ \begin{bmatrix} 1 & 1 & 1 \\ 2 & 4 & 3 \\ 5 & 10 & 6 \end{bmatrix} & \begin{bmatrix} 70 \\ 88 \\ 78 \end{bmatrix} \end{matrix}$$

$$\begin{bmatrix} 236 \\ 726 \\ 1698 \end{bmatrix} \begin{matrix} 1(70) + 1(88) + 1(78) \\ 2(70) + 4(88) + 3(78) \\ 5(70) + 10(88) + 6(78) \end{matrix}$$

$$X\beta = 0$$

$$\nabla J(0) = \frac{1}{3} (0 - Y)^T X$$

$$= \frac{1}{3} X^T Y$$

$$= \frac{1}{3} \begin{bmatrix} 236 \\ 726 \\ 1698 \end{bmatrix} = \begin{bmatrix} -78.67 \\ -242 \\ -566 \end{bmatrix}$$

Update $\beta \leftarrow \beta - \alpha \nabla J(\beta)$ $\alpha = 0.001$

$$\beta_{\text{new}} = -0.001 \times \nabla J(0) = \begin{bmatrix} 0.0787 \\ 0.242 \\ 0.566 \end{bmatrix}$$

Q.15 Multiple Linear Regression

X_1 (TV) X_2 (Radio) Y (Sales)

1	1	6
2	0	7
0	2	5
3	1	10

→ Model $\hat{y} = \beta_0 + \beta_1 X_1 + \beta_2 X_2$

→ Matrix Form

$$X = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 0 \\ 1 & 0 & 2 \\ 1 & 3 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 6 \\ 7 \\ 5 \\ 10 \end{bmatrix}$$

→ Normal Equation $B = (X^T X)^{-1} X^T Y$

$$X^T = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 2 & 0 & 3 \\ 1 & 0 & 2 & 1 \end{bmatrix} \quad X^T X = \begin{bmatrix} 4 & 6 & 4 \\ 6 & 14 & 4 \\ 4 & 4 & 6 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 28 \\ 50 \\ 28 \end{bmatrix} \quad \begin{aligned} &1(6) + 1(7) + 1(5) + 1(10) \\ &1(6) + 2(7) + 0(5) + 3(10) \\ &1(6) + 0(7) + 2(5) + 1(10) \end{aligned}$$

$$\begin{bmatrix} 4 & 6 & 4 \\ 6 & 14 & 4 \\ 4 & 4 & 6 \end{bmatrix}$$

changed matrix algorithm 21.2

$$(102)^T (0.0007, X) (0.0007, X)$$

2

1

1

f

0

5

2

5

0

0

1

5

$$1 + 1 + 1 + 1 = 4$$

$$1(1) + 1(2) + 0 + 3 = 6$$

$$1 + 0 + 2 + 1 = 4$$

$$1(1) + 1(2) + 0(3) + 3(4) = 10 \text{ leaves } \leftarrow$$

$$\begin{bmatrix} 2 \\ f \\ 2 \\ 0 \end{bmatrix} = P$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 5 & 1 \\ 2 & 0 & 1 \\ 1 & 5 & 1 \end{bmatrix} = X$$

matrix X

$$P^T X^T (X^T X)^{-1} X^T \text{ normal equation } \leftarrow$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 1 & 1 & 1 \\ 2 & 1 & 1 \end{bmatrix} = X^T X$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 5 & 0 & 5 & 1 \\ 1 & 5 & 0 & 1 \end{bmatrix} = X^T X$$

$$\begin{bmatrix} 1(1) + 1(1) + 1(5) + 1(1) & 1(2) + 1(1) + 1(5) + 1(1) \\ 1(2) + 1(1) + 1(5) + 1(1) & 1(1) + 1(2) + 1(5) + 1(1) \\ 1(1) + 1(2) + 1(5) + 1(1) & 1(1) + 1(1) + 1(5) + 1(1) \end{bmatrix} = X^T X$$

Finding Inverse

$$\rightarrow \begin{bmatrix} 4 & 6 & 4 \\ 6 & 14 & 4 \\ 4 & 4 & 6 \end{bmatrix} \det \begin{bmatrix} 4(14 \times 6 - (4 \times 4)) - 6 \\ (6 \times 6) - (4 \times 4) + 4 \\ (6 \times 4) - (4 \times 4) \end{bmatrix}$$

$$4(84 - 16) - 6(36 - 16) + 4(-32)$$

$$272 - 120 - 128 = 24$$

Cofactors

$$\begin{array}{lcl} u = 68 & , & 6 = -20 & , & 4 = -32 \\ 6 = -20 & , & 14 = 8 & , & 4 = +8 \\ 4 = -32 & , & 4 = 8 & , & 6 = 20 \end{array}$$

$$\begin{bmatrix} 68 & -20 & -32 \\ -20 & 8 & 8 \\ -32 & 8 & 20 \end{bmatrix}$$

$\rightarrow$ Adjoint, Transpose of cofactor matrix

$$\begin{bmatrix} 68 & -20 & -32 \\ -20 & 8 & 8 \\ -32 & 8 & 20 \end{bmatrix} \text{ Inverse } \frac{1}{24} \begin{bmatrix} 68 & -20 & -32 \\ -20 & 8 & 8 \\ -32 & 8 & 20 \end{bmatrix}$$

$$\begin{bmatrix} 2.833 & -0.833 & -1.333 \\ -0.833 & 0.333 & 0.333 \\ -1.333 & 0.333 & 0.833 \end{bmatrix} \begin{bmatrix} 28 \\ 50 \\ 26 \end{bmatrix}$$

$$P_0 = (2.833)(28)$$

$$79.324 - 41.65 - 34.658 = 3$$

20

amount paid?

$$2 - (1768 \times (28 \times 100)) 1904 \text{ up}$$

$$N = 20 \times 100 (2 \times 2) = 1000$$

$$[(1768 \times 32 \times 26 \times 2) - 832] \begin{bmatrix} N & 2 & N \\ N & 72 & 2 \\ 2 & 2 & N \end{bmatrix} \leftarrow$$

$$(28 - 1)N + (26 - 25)2 - (21 - 19)N$$

$$(28 - 1)N + 2 - 2 = 28N$$

amount

$$\begin{matrix} 28 - 1 = N & 26 - 25 = 2 & 21 - 19 = N \\ 8 + 1 = N & 8 = N & 28 - 2 = N \\ 28 = 2 & 8 = N & 28 = N \end{matrix}$$

$$\begin{bmatrix} 28 - 1 & 26 - 25 & 21 - 19 \\ 8 & 8 & 28 \\ 28 & 8 & 28 \end{bmatrix}$$

matrix inverse for logarithm - inverse

$$\begin{bmatrix} 28 - 1 & 26 - 25 & 21 - 19 \\ 8 & 8 & 28 \\ 28 & 8 & 28 \end{bmatrix} \frac{1}{28}$$

$$(X^T X)^{-1} = \begin{bmatrix} 1.333 & 0.833 & 0.833 \\ 0.833 & 1.333 & 0.833 \\ 0.833 & 0.833 & 1.333 \end{bmatrix}$$

$$\begin{bmatrix} 28 \\ 28 \\ 28 \end{bmatrix}$$

$$E = 880.1N - 72.1N - 28.2N$$

$$\beta_1 = \frac{(-20)(28) + 8(80) + 8(26)}{-560 + 400 + 208} = \frac{48}{24} = 2$$

$$\beta_2 = \frac{(-32)(28) + 8(80) + (20)(26)}{-896 + 400 + 520} = \frac{24}{24} = 1$$

$$\begin{bmatrix} 3 \\ 2 \\ 1 \end{bmatrix}$$

$$\rightarrow \hat{y} = 3 + 2x_1 + 1x_2$$

$$\rightarrow \text{Predicting Sales } x_1 = 4 \quad x_2 = 2$$

$$3 + 2(4) + 2 = 3 + 8 + 2 = 13$$

(15) Underfitting v Overfitting

$$\rightarrow \text{Model A: } \hat{y} = \beta_0 + \beta_1 x$$

$$\text{Model B: } \hat{y} = \beta_0 + \beta_1 x + \beta_2 x^2 \dots \beta_n x^n$$

Model	Train	Test Error
	MSE	MSE
A	12	15
B	0.5	50

$\rightarrow$ Which model is Underfitting

Model A Train E = 12 Test E = 15

Both errors are high and Simple

Model is too simple, Cannot capture pattern

→ Overfitting

Model B : T.E = 0.5 Extremely Small
 $y_i \approx \hat{y}_i$ So model fits training data almost perfectly.

Test Error is Very large

D/D Test vs Training is 0.5 - 0.5, 0.49.5

→ This means Model performs well on Training but poorly on new data

→ This model learns noise and real patterns
 So instead of learning true relationship it memorizes every training point.

→ If you wanted better model, the degree of polynomial you might try? Why?

Ans

$$Y = \beta_0 + \beta_1 X + \beta_2 X^2 \text{ or } \beta_3 X^3$$

This would balance bias and Variance, giving lower test error

16 Bias - Variance Trade off.

→ We train model on some dataset. Expected Squared Error at test point

$$E(Y - f(X))^2 = \text{Bias}^2 + \text{Variance} + \sigma^2$$

σ^2 is irreducible noise

Model	Bias ²	Variance	Noise (σ^2)
Simple linear	25	4	9
Complex Polynomial	4	36	9

→ Compute Total Expected Error

$$\begin{aligned} S \quad E &= 25 + 4 + 9 = 38 \\ C &= 4 + 36 + 9 = 49 \end{aligned}$$

→ Which model has higher bias?

Simple model has higher bias $25 > 4$
Cannot capture complex patterns.

→ Higher Variance $36 > 4$ Complex

→ Simple linear model is preferred since $38 < 49$

It has low total Expected Error

(17) Regularization : Ridge V Lasso

To avoid overfitting and handle multicollinearity we consider regularization

→ Optimization Problem of Ridge Regression
It adds L2 penalty to loss function.

$$\min_{\beta} \left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p \beta_j^2 \right]$$

β_j : Coefficients λ = Regularization Param ^{Prediction} _{Penalty}

Loss : Sum of Squared Error + penalty on Squared coefficients

→ Lasso Regularization : It adds L1 penalty.

$$\min_{\beta} \left[\sum_{i=1}^n (y_i - \hat{y}_i)^2 + \lambda \sum_{j=1}^p |\beta_j| \right]$$

Loss : Sum of Squared Errors + absolute coefficient ^{Penalty}

→ Difference in Effect on Coefficients.
penalty $\{ \beta_j^2 \}$ (Shrink towards zero, Never make them exactly zero)

Lasso $\{ \beta_j \}$ Shrink Coefficient Some become exactly zero so performs feature Selection

→ When Ridge is Preferred.

- ① When many predictors are correlated
- ② When all predictors are useful and should remain in model

→ When Lasso is Preferred

- Automatic feature Selection Eg Important genes in medical
- When dataset has many irrelevant predictors
Eg Text classification with thousand of features

→ Example of Ridge

- ① Economic indicators predicting GDP.
- ② Sensor data where each variables have information

→ Effects of Regularization Parameter λ

λ controls strength of Regularization If $\lambda = 0$
No Regularization

→ If λ increases coefficient shrink it becomes simple

→ Bias Variance Effect Increasing λ (Bias Increases)
Variance decreases) Simpler model less sensitive
to training data.

(18) Numerical Example : Ridge

$$X = \begin{bmatrix} 1 & 0 \\ 0 & 1 \\ 1 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}$$

$$X^T = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 1 \end{bmatrix} \quad X^T X = \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}$$

$$X^T Y = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

→ Ordinary least Square $\beta_{OLS} = (X^T X)^{-1} X^T Y$

deter $X^T X = (2 \times 2) - (1 \times 1) = 4 - 1 = 3$

$$(X^T X)^{-1} = \frac{1}{3} \begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix} \quad \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

$$\begin{aligned} 2(3) + (-1)(3) &= 6 - 3 = 3/3 = 1 \\ -1(3) + 2(3) &= -3 + 6 = 3/3 = 1 \end{aligned} \quad \beta_{OLS} = \begin{bmatrix} 1 \\ 1 \end{bmatrix}$$

→ Ridge $(X^T X + \lambda I)^{-1} X^T Y \quad \lambda = 1$

$X^T X + \lambda I$ (Identity mat)

$$I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\rightarrow \text{Inverse of Ridge} = \begin{bmatrix} 3 & 1 \\ 1 & 3 \end{bmatrix}$$

$$\det(A) = 9 - 1 = 8$$

$$\frac{1}{8} \begin{bmatrix} 3 & -1 \\ -1 & 3 \end{bmatrix} \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$9 - 1 = \frac{6}{8}$$

$$-3 + 9 = \frac{6}{8}$$

$$\begin{bmatrix} 0.75 \\ 0.75 \end{bmatrix}$$

$\rightarrow$ Shrinkage Effect at shrinkage coefficient towards zero $1 \rightarrow 0.75$ become penalty Term discourages large coefficient